B.1 Lab 8: Planning a Thermal Ablation

The Planning Problem

Chapter 7 opened with laser interstitial thermal therapy: a thin fiber is guided to the center of a brain tumor, the tip heats the surrounding tissue, and the treatment succeeds when every point of the tumor is held at the cell-killing temperature of 43 degrees Celsius while the healthy tissue nearby is spared (Figure 7.1). In this lab, you will compute an actual treatment plan: given the tumor geometry, you will determine where to place the fiber and how much power to deliver.

The model needs one physical ingredient beyond Chapter 7. A steady state requires a heat sink. In living tissue, the dominant sink is blood perfusion. Blood arrives at arterial temperature, absorbs heat, and carries it away, and the brain is one of the most heavily perfused organs in the body. Pennes modeled this effect as a cooling term proportional to the difference between the tissue temperature and the arterial temperature [12], and his bioheat equation remains the standard model for thermal therapy planning. Writing $w(x, y)$ for the temperature elevation above body temperature, so that $u = 37 + w$, the steady state bioheat equation on our normalized domain is

$$-\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}\right) + \beta\, w = q(x, y),$$

where $q(x, y)$ is the absorbed power density supplied by the laser, the rate at which the fiber deposits heat per unit area of tissue, and the constant β lumps the perfusion rate and the heat capacity of blood into a single normalized number. In physical units, $\beta = \omega \rho_b c_b L^2/k$, where ω is the volumetric blood perfusion rate, $\rho_b c_b$ is the volumetric heat capacity of blood, k is the thermal conductivity of the tissue, and L is the width of the domain. The quantity $1/\sqrt{\beta}$ is the distance, in domain units, over which perfusion damps out a temperature disturbance. We will use $\beta = 50$ throughout this lab, which corresponds to brain-like perfusion on a domain roughly three centimeters across.

Discretizing with the five-point stencil of Chapter 7 converts the equation into a linear system. The conduction term produces the matrix A that `laplacianMat` builds. The boundary conditions are zero because the boundary rests at body temperature, and the perfusion term adds β to every diagonal entry. The right-hand side $\mathbf{s}$ holds the discretized laser source rather than boundary data, which means that it is zero at every node except the one holding the fiber tip.

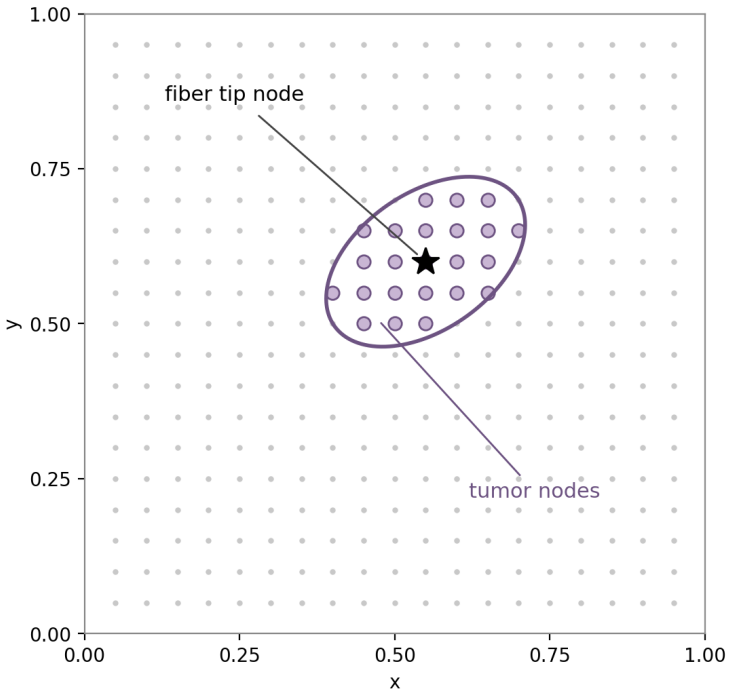
$$(A + \beta I)\,\mathbf{w} = \mathbf{s}.$$

The fiber tip occupies a single grid node. For the single fiber plan of Problems 1 and 2 the tip sits at the center $(0.55, 0.60)$ of the ellipse defined below, and Problem 3 moves it. A tip delivering total power P into the area h^2 surrounding its node corresponds to a power density of P/h^2 at that node and zero everywhere else, where $h = 0.05$ is the grid spacing. Because the system is linear, the field produced by power P is P times the field produced by unit power, and writing $\mathbf{v}$ for the unit power solution gives $\mathbf{w} = P\,\mathbf{v}$; this is why Problem 1 solves for $\mathbf{v}$ once and Problem 2 only scales it.

The tumor is the rotated ellipse shown below. A grid point (x, y) lies inside the tumor when

$$\left(\frac{\bar{x}}{0.18}\right)^2 + \left(\frac{\bar{y}}{0.11}\right)^2 \le 1, \qquad \begin{aligned} \bar{x} &= \cos(\theta)(x - 0.55) + \sin(\theta)(y - 0.60) \\ \bar{y} &= -\sin(\theta)(x - 0.55) + \cos(\theta)(y - 0.60) \end{aligned}$$

with $\theta = 35$ degrees. On the 19×19 interior grid, 23 nodes fall inside the tumor.



Problem 1: The Unit Power Field

Write a function `unitField.m` with the calling sequence

```
[v, D] = unitField(beta, tip)
```

where `tip` is a vector containing the (x, y) coordinates of the fiber tip. The function builds the system matrix using `[A, b, D] = laplacianMat([0 0 0 0])` from Lab 6, Section B.6, adds the perfusion term, places a unit power source at the grid node nearest to `tip`, and solves for the resulting temperature elevation field $\mathbf{v}$ with backslash. Recall from Lab 6 that the solution vector reshapes onto the grid with `V(:) = v` and plots with `surf(D, D, V)`.

Create a `surf` plot of the unit power field for $\beta = 50$ with the tip at $(0.55, 0.60)$. The elevation at the tip node should be 0.4353 for $\beta = 50$ and 0.6287 for $\beta = 0$, to the digits shown. In your header, explain what the term $\beta\mathbf{w}$ models physically and what happens to the solution when β is set to zero.

Problem 2: The Minimum Power

The system $(A + \beta I)\mathbf{w} = \mathbf{s}$ is linear, and the source vector is proportional to the power P. Use this structure to determine, from your single unit power solve and no additional system solves, the minimum power $P_{\min}$ at which every tumor node reaches 43 degrees, that is, $w \geq 6$ at every tumor node. State your reasoning in the header of a script `minPower.m` that carries out the computation.

Your script should build the tumor mask from the ellipse inequality above, compute $P_{\min}$ for $\beta = 50$ and for $\beta = 0$, and display both values along with the temperature at the tip node for each. As a validation check, $P_{\min} = 107.94$ for $\beta = 50$, to the digits shown. For each value of β, plot the 43 degree contour of the resulting temperature field over the tumor boundary using `contour(D, D, U, [43 43])`, where `U = 37 + V` is the temperature on the grid. In your header, answer the planning question these results raise: identify what cools the tissue in each of the two cases, and explain why perfusion more than triples the power required to cover the tumor.

Problem 3: Planning with Two Fibers

Clinical LITT treatments of elongated tumors often use more than one heating position. The fiber is advanced or withdrawn along its track and fired at multiple positions. Our tumor is elongated, and the model should show the benefit.

Suppose two fiber tips are placed at grid nodes k_1 and k_2, each delivering the same power P. By linearity, the resulting elevation field is $P(\mathbf{v}_{k_1} + \mathbf{v}_{k_2})$, where $\mathbf{v}_k$ denotes the unit power field for a tip at node k. Write a script `planTwoFibers.m` that searches every pair of tumor nodes as candidate tip positions, computes for each pair the minimum per fiber power that covers the tumor, and selects the pair minimizing the total power $2P$. For the winning plan and for the single fiber plan of Problem 2, your script should display the total power, the peak tissue temperature, and the number of healthy nodes at or above 43 degrees. The winning total power is 88.70, to the digits shown.

With 23 tumor nodes there are 253 pairs, and each candidate tip needs one unit power solve, and the script should solve each tumor node's unit field once before the pair loop. In your header, report all three comparisons and explain why the two fiber plan wins on every one of them for a tumor of this shape.

One caveat is in order. The temperature at the tip node itself should not be trusted, because the model concentrates the entire power into one node of area h^2. A true point source in two dimensions produces a temperature that grows like the logarithm of the distance from it, which means that the tip value climbs without bound as the grid is refined, while the temperatures a few nodes away converge normally. The tumor coverage and the healthy node counts, which depend on those surrounding values, are the meaningful outputs of this model. Comparisons between plans on the same grid, like the peak temperatures of Problem 3, remain fair, because both plans carry the same bias. Real treatment planning systems solve the transient Pennes equation with patient specific perfusion maps and track the accumulated thermal dose over the course of the treatment, but the structure of the calculation is the one you just performed.

General Submission Guidelines

```
[FirstName][LastName]_[filename].m           For programs
[FirstName][LastName]_[filename]_plot_1.pdf  For corresponding plots (export them to pdf)
[FirstName][LastName]_[filename]_plot_2.pdf  For problem with more than one plot
```

Submission Checklist

- Problem 1: `unitField.m`
  - Fully commented code and complete header
  - Correct system assembly with the perfusion term and the unit source
  - Surf plot of the unit power field matching the validation values
  - Header explanation of the perfusion term
- Problem 2: `minPower.m`
  - Header statement of the minimum power reasoning
  - Correct tumor mask and both $P_{\min}$ values displayed
  - Contour plots for both values of β with the tumor boundary

  - Header explanation of the perfusion comparison

- Problem 3: `planTwoFibers.m`
  - Correct pair search with unit fields solved once per tumor node
  - Total power, peak temperature, and healthy node counts for both plans
  - Header explanation of why the two fiber plan wins